

Lateral Forces

SUBMITTED TO: DR. ABDEL HAMID ZAGHW

Submitted By

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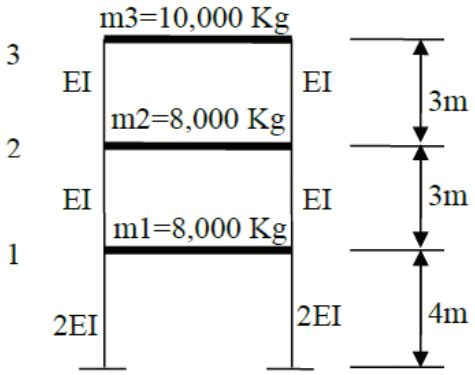
Sec: 2

B.N: 1

2023 -2024

Factor = Sec + BN/100 = 2+1/100 = 2.01.

Q1- Mass and Stiffness Matrices.



$m1 = 8000 \times 2.01 = 16080 = m2.$

$m3 = 10000 \times 2.01 = 20100.$

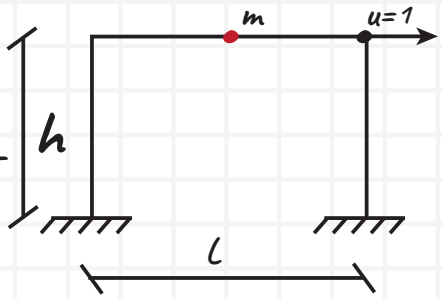
$$[M] = \begin{bmatrix} 16080 & 0 & 0 \\ 0 & 16080 & 0 \\ 0 & 0 & 20100 \end{bmatrix}$$

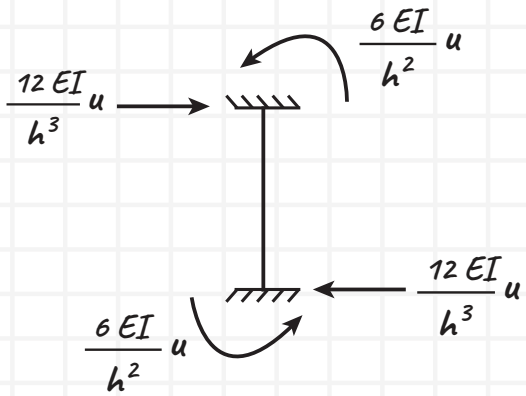
$EI = 36{,}000\ KN.m^2$

$EI = 36 \times 10^6\ N.m^2$

* To Find Stiffness Matrics.

$$K = \frac{24\,EI_c}{h^3} + h$$



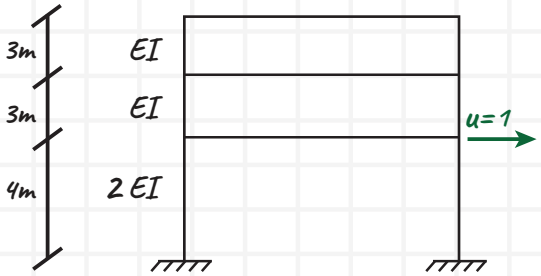


* To Find Stiffness Matrics.

u = 1 @ 1st Floor.

$$K_{11} = \left(\frac{24 \times 72 \times 10^6}{4^3} + \frac{24 \times 36 \times 10^6}{3^3} \right)$$

$$= 59 \times 10^6\ N/m.$$



$$K_{21} = \left(\frac{-24 \times 36 \times 10^6}{3^3} \right) = -32 \times 10^6\ N/m.$$

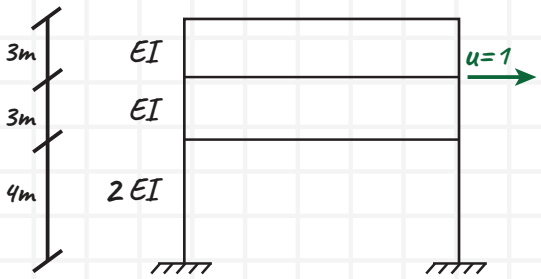
K_{31} = Zero.

u = 1 @ 2nd Floor.

$$K_{12} = \left(\frac{-24 \times 36 \times 10^6}{3^3}\right) = -32 \times 10^6 \ N/m.$$

$$K_{22} = \left(\frac{24 \times 36 \times 10^6}{3^3}\right) \times 2 = 64 \times 10^6 \ N/m.$$

$$K_{32} = \left(\frac{-24 \times 36 \times 10^6}{3^3}\right) = -32 \times 10^6 \ N/m.$$

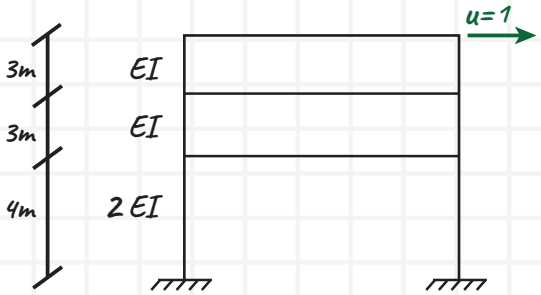


u = 1 @ 3rd Floor.

K_{13} = Zero.

$$K_{23} = \left(\frac{-24 \times 36 \times 10^6}{3^3}\right) = -32 \times 10^6 \ N/m.$$

$$K_{33} = \left(\frac{24 \times 36 \times 10^6}{3^3}\right) = 32 \times 10^6 \ N/m.$$



* So stiffnes Matrics is.

$$[K] = \begin{bmatrix} 59 & -32 & 0 \\ -32 & 64 & -32 \\ 0 & -32 & 32 \end{bmatrix} \times 10^6 \ N/m.$$

Q2-To find natural frequecy. (Eigen Value)

$$[K] - W^2 [M] = 0 \longrightarrow \text{Let } W^2 = \lambda$$

$$\begin{bmatrix} 59 & -32 & 0 \\ -32 & 64 & -32 \\ 0 & -32 & 32 \end{bmatrix} \times 10^6 - W^2 \begin{bmatrix} 16080 & 0 & 0 \\ 0 & 16080 & 0 \\ 0 & 0 & 20100 \end{bmatrix} = 0$$

$$\begin{vmatrix} +(59\times10^6-16080\lambda) & -(-32\times10^6) & +(0) \\ -32\times10^6 & +(64\times10^6-16080\lambda) & -(-32\times10^6) \\ 0 & -32\times10^6 & +(32\times10^6-20100\lambda) \end{vmatrix} = 0$$

$$+(59\times10^6-16080\lambda)\left[(64\times10^6-16080\lambda)(32\times10^6-20100\lambda)-(-32\times10^6)(-32\times10^6)\right]$$

$$-(64\times10^6-16080\lambda)\left[(-32\times10^6)(32\times10^6-20100\lambda)-0\right]$$

$$+(32\times10^6-20100\lambda)\left[(59\times10^6-16080\lambda)(64\times10^6-16080\lambda)-(-32\times10^6)(-32\times10^6)\right]$$

$$=\left[-5.19718464\times10^{12}\lambda^3+4.80287088\times10^{16}\lambda^2-1.2272256\times10^{12}\lambda+6.0416\times10^{22}\right]$$

$$+\left[-5.19718464\times10^{12}\lambda^3-4.80287088\lambda^2+1.3918848\times10^{20}\lambda-1.20832\times10^{23}\right]$$

$$+\left[-5.19718464\times10^{12}\lambda^3+4.80287088\lambda^2-1.1860608\times10^{20}\lambda+8.8064\times10^{22}\right]$$

$$=\left[-5.19718464\times10^{12}\lambda^3+4.80287088\times10^{16}\lambda^2-1.0214016\times10^{20}\lambda+2.7648\times10^{22}\right]$$

$\therefore \lambda_1 = 6218.42285 \longrightarrow W^2 = 78.85697718$ rad/sec

$\therefore \lambda_2 = 2706.820647 \longrightarrow W^2 = 52.02711454$ rad/sec

$\therefore \lambda_3 = 316.0500355 \longrightarrow W^2 = 17.77779614$ rad/sec

* To find Mode Shapes. (Eigen Vector)

For $\lambda_1 = 6218.42285$

$$\begin{bmatrix} -40992239.43 & -32\times10^6 & 0 \\ -32\times10^6 & -35992239.43 & -32\times10^6 \\ 0 & -32\times10^6 & -92990299.29 \end{bmatrix} \begin{bmatrix} \varnothing_{11} \\ \varnothing_{21} \\ \varnothing_{31} \end{bmatrix}$$

$$-40992239.43\,\varnothing_{11} - 32\times10^6\,\varnothing_{21} = 0$$

Assume $\varnothing_{11} = 1 \longrightarrow \varnothing_{21} = -1.281007482$

$$-32\times10^6\,\varnothing_{21} - 92990299.29\,\varnothing_{31} = 0.4408227497$$

$$\begin{bmatrix} 1 \\ -1.281007482 \\ 0.4408227497 \end{bmatrix}$$

For $\lambda_2 = 2706.820647$

$$\begin{bmatrix} 15474324 & -32\times10^6 & 0 \\ -32\times10^6 & 20474324 & -32\times10^6 \\ 0 & -32\times10^6 & -22407095 \end{bmatrix} \begin{bmatrix} \varnothing_{11} \\ \varnothing_{21} \\ \varnothing_{31} \end{bmatrix}$$

$$15474324\,\varnothing_{11} - 32\times10^6\,\varnothing_{21} = 0$$

Assume $\varnothing_{11} = 1 \longrightarrow \varnothing_{21} = 0.483572625$

$$-32\times10^6\,\varnothing_{21} - 22407095\,\varnothing_{31} = -0.6905992946$$

$$\begin{bmatrix} 1 \\ 0.483572625 \\ -0.6905992946 \end{bmatrix}$$

For $\lambda_2 = 2706.820647$

$$\begin{bmatrix} 53917915.43 & -32\times10^6 & 0 \\ -32\times10^6 & 58917915.43 & -32\times10^6 \\ 0 & -32\times10^6 & 25647394.29 \end{bmatrix} \begin{bmatrix} \varnothing_{11} \\ \varnothing_{21} \\ \varnothing_{31} \end{bmatrix}$$

$$53917915.43\,\varnothing_{11} - 32\times10^6\,\varnothing_{21} = 0$$

Assume $\varnothing_{11} = 1 \longrightarrow \varnothing_{21} = 1.684934857$

$$-32\times10^6\,\varnothing_{21} + 25647394.29\,\varnothing_{31} = 2.102276544$$

$$\begin{bmatrix} 1 \\ 1.684934857 \\ 2.102276544 \end{bmatrix}$$

Q3-To find Modal Mass .

$$m_i = \{\varnothing_i\}^T [M] \{\varnothing_i\} = \sum_{i=1}^{n} m_{jj} (\varnothing_i^2)_{mode\,i}$$

For mode 1

$$m_1 = \begin{bmatrix} 1 & -1.281007482 & 0.4408227497 \end{bmatrix} \begin{bmatrix} 16080 & 0 & 0 \\ 0 & 16080 & 0 \\ 0 & 0 & 20100 \end{bmatrix} \begin{bmatrix} 1 \\ -1.281007482 \\ 0.4408227497 \end{bmatrix}$$

$$= \boxed{46372.88752}$$

Check.

$$m_1 = 16080(1)^2 + 16080(-1.281007482)^2 + 20100(0.4408227497)^2 = \boxed{46372.88752}$$ ✓

For mode 2

$$m_2 = 16080(1)^2 + 16080(0.483572625)^2 + 20100(-0.6905992946)^2 = \boxed{19840.18711}$$

For mode 3

$$m_3 = 16080(1)^2 + 16080(-1.281007482)^2 + 20100(0.4408227497)^2 = \boxed{150564.498}$$

Q4-To find Normalized Mode Shapes .

For mode 1

$$\begin{Bmatrix} \emptyset_{11} \\ \emptyset_{21} \\ \emptyset_{31} \end{Bmatrix} = \begin{Bmatrix} 1 \\ -1.281007482 \\ 0.4408227497 \end{Bmatrix} \xrightarrow{\text{Normalizing}} \begin{Bmatrix} \frac{1}{\sqrt{46372.88752}} \\ \frac{-1.281007482}{\sqrt{46372.88752}} \\ \frac{0.4408227497}{\sqrt{46372.88752}} \end{Bmatrix} = \begin{Bmatrix} 4.643740369 \times 10^{-3} \\ -5.948666157 \times 10^{-3} \\ 2.047066399 \times 10^{-3} \end{Bmatrix}$$

For mode 2

$$\begin{Bmatrix} \emptyset_{11} \\ \emptyset_{21} \\ \emptyset_{31} \end{Bmatrix} = \begin{Bmatrix} 1 \\ 0.483572625 \\ -0.6905992946 \end{Bmatrix} \xrightarrow{\text{Normalizing}} \begin{Bmatrix} \frac{1}{\sqrt{19840.18711}} \\ \frac{0.483572625}{\sqrt{19840.18711}} \\ \frac{-0.6905992946}{\sqrt{19840.18711}} \end{Bmatrix} = \begin{Bmatrix} 7.099489451 \times 10^{-3} \\ 3.43311875 \times 10^{-3} \\ -4.902902407 \times 10^{-3} \end{Bmatrix}$$

For mode 3

$$\begin{Bmatrix} \emptyset_{11} \\ \emptyset_{21} \\ \emptyset_{31} \end{Bmatrix} = \begin{Bmatrix} 1 \\ 1.684934857 \\ 2.102276544 \end{Bmatrix} \xrightarrow{\text{Normalizing}} \begin{Bmatrix} \frac{1}{\sqrt{150564.498}} \\ \frac{1.684934857}{\sqrt{150564.498}} \\ \frac{2.102276544}{\sqrt{150564.498}} \end{Bmatrix} = \begin{Bmatrix} 2.577144142 \times 10^{-3} \\ 4.342319997 \times 10^{-3} \\ 5.417869681 \times 10^{-3} \end{Bmatrix}$$

Q5-To find Orthonormal Modal Matrix .

$$[A] = \begin{bmatrix} 4.643740369 & 7.099489451 & 2.577144142 \\ -5.948666157 & 3.43311875 & 4.342319997 \\ 2.047066399 & -4.902902407 & 5.417869681 \end{bmatrix} \times 10^{3}$$

Q6-To find the Participation Factors.

$$\Gamma_i = \frac{m_i(\emptyset_{jj}) + m_i(\emptyset_{jj}) + m_i(\emptyset_{jj})}{m_i(\emptyset_{jj}^2) + m_i(\emptyset_{jj}^2) + m_i(\emptyset_{jj}^2)}$$

For mode 1

$$\Gamma_1 = \frac{(46372.88752 \times 1) + (19840.8711 \times -1.281007482) + (150564.498 \times 0.4408227497)}{(46372.88752 \times 1^2) + (19840.8711 \times (-1.281007482)^2) + (150564.498 \times (0.4408227497)^2)}$$

$$= \boxed{0.8071820848}$$

For mode 2

$$\Gamma_1 = \frac{(46372.88752 \times 1) + (19840.8711 \times 0.483572625) + (150564.498 \times -0.6905992946)}{(46372.88752 \times 1^2) + (19840.8711 \times (0.483572625)^2) + (150564.498 \times (-0.6905992946)^2)}$$

$$= \boxed{-0.3909136205}$$

For mode 3

$$\Gamma_1 = \frac{(46372.88752 \times 1) + (19840.8711 \times 1.684934857) + (150564.498 \times 2.102276544)}{(46372.88752 \times 1^2) + (19840.8711 \times (1.684934857)^2) + (150564.498 \times (2.102276544)^2)}$$

$$= \boxed{0.5159688144}$$

Q7-To find Modal Weight(Mass) ratio. (mass participation factor)

$$MMR = \frac{(\sum m_{jj} \varnothing_j)^2}{(\sum m_{jj} \varnothing_j)(\sum m_{jj})}$$

$$MMR_1 = \frac{(16080x1+16080x(-1.281007482)+20100x0.4408227497)}{(46372.88752)(16080+16080+20100)}$$

$$MMR_1 = 7.779173945x10^{-3}$$

$$MMR_2 = \frac{(16080x1+16080x(-1.281007482)+20100x-(0.6905992946))}{(46372.88752)(16080+16080+20100)}$$

$$MMR_2 = 0.54887574x10^{-3}$$

$$MMR_3 = \frac{(16080x1+16080x1.684934857+20100x2.102276544)}{(150564.498)(16080+16080+20100)}$$

$$MMR_3 = 0.9275212224x10^{-3}$$

Q8-To find shear at each level & The total base shear due to each mode.

For mode 1 $S_a = 0.1g = 0.1x9.81 = 0.981$ & $\Gamma_i x S_a = 0.7918617852$

$$\begin{bmatrix} P_1 \\ P_2 \\ P_3 \end{bmatrix} = \begin{bmatrix} 16080 & 0 & 0 \\ 0 & 16080 & 0 \\ 0 & 0 & 20100 \end{bmatrix} \begin{bmatrix} 1 \\ -1.281007482 \\ 0.4408227497 \end{bmatrix} x\ 0.7918617852 = \begin{bmatrix} 101865.1 \\ -130489.9553 \\ 56130.56687 \end{bmatrix} N$$

For mode 2 $S_a = 0.2g = 0.2x9.81 = 1.962$ & $\Gamma_i x S_a = -0.7669725234$

$$\begin{bmatrix} P_1 \\ P_2 \\ P_3 \end{bmatrix} = \begin{bmatrix} 16080 & 0 & 0 \\ 0 & 16080 & 0 \\ 0 & 0 & 20100 \end{bmatrix} \begin{bmatrix} 1 \\ 0.483572625 \\ -0.6905992946 \end{bmatrix} x\ -0.7668725234 = \begin{bmatrix} 34613.1606 \\ 16737.97687 \\ -0.02987978017 \end{bmatrix} N$$

For mode 3 $S_a = 0.3g = 0.3x9.81 = 2.943$ & $\Gamma_i x S_a = 1.415302458$

$$\begin{bmatrix} P_1 \\ P_2 \\ P_3 \end{bmatrix} = \begin{bmatrix} 16080 & 0 & 0 \\ 0 & 16080 & 0 \\ 0 & 0 & 20100 \end{bmatrix} \begin{bmatrix} 1 \\ 1.684934857 \\ 2.102276544 \end{bmatrix} x\ 1.415302458 = \begin{bmatrix} 24417.40995 \\ 41141.74545 \\ 64165.18527 \end{bmatrix} N$$

To get total Base Shear. $(V_i)_{max} = m_1 \times \Gamma_i^2 \times S_a$

For mode 1

$$(V_i)_{max} = 46372.88752 \times 0.8071820848 \times 0.981 = 29641.07164\ N$$

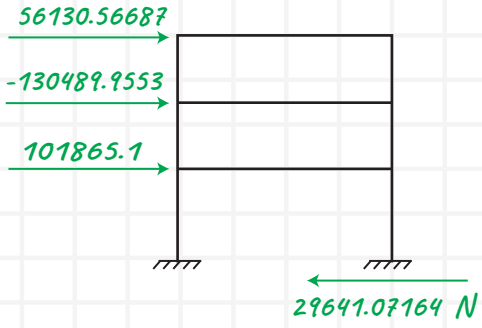
For mode 2

$$(V_i)_{max} = 19840.18711 \times -0.3909136205 \times 1.962 = 46855.13424\ N$$

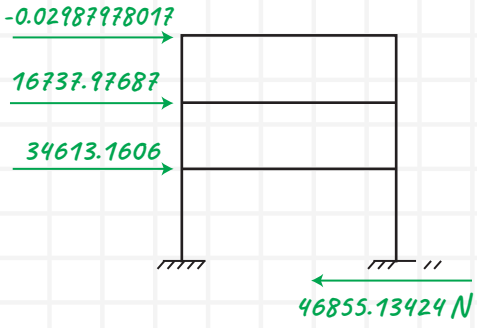
For mode 3

$$(V_i)_{max} = 150564.498 \times 0.5159688144 \times 2.943 = 117966.6969\ N$$

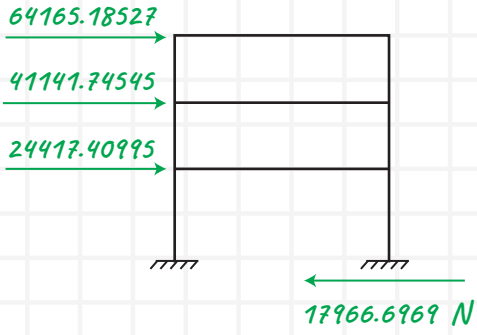
For mode 1



For mode 2



For mode 3



Q9-To find shear at each level & The total base shear by using square root of the sum of the squares .

$$P_1 = \sqrt{(56130.56687)^2 + (-0.02987978017)^2 + (64165.18527)^2} = 85251.46062$$

$$P_2 = \sqrt{(-130489.9553)^2 + (16737.97687)^2 + (41141.74515)^2} = 137842.0527$$

$$P_3 = \sqrt{(101865.1)^2 + (34613.1606)^2 + (24417.40995)^2} = 110321.2254$$

$$V_{total} = P_1 + P_2 + P_3 = 33414.7687$$

